1 Introduction

Falls represent a critical health concern for older adults, accounting for over 38 million medically-treated injuries annually worldwide [1]. Wearable fall detection systems offer a promising solution for continuous, non-intrusive monitoring using commodity devices such as smartwatches. However, developing accurate fall detection models that operate within the computational and power constraints of wearable devices remains challenging.

Modern smartwatches contain inertial measurement units (IMUs) comprising triaxial accelerometers and gyroscopes. While accelerometers measure linear acceleration and provide reliable impact detection, gyroscopes measure angular velocity and capture rotational motion patterns characteristic of falls. Intuitively, combining both modalities should improve detection performance. However, our preliminary investigation revealed a counterintuitive finding: adding raw gyroscope data to accelerometer-only models *degraded* performance by 1.6–2.0% F1 score.

Analysis revealed that single-stream architectures, which process concatenated accelerometer and gyroscope channels through shared convolutional layers, allow noisy gyroscope signals to corrupt discriminative accelerometer features. Consumer-grade MEMS gyroscopes exhibit substantial measurement noise and bias drift, with signal-to-noise ratios below 1.0 for over 75% of subjects in our dataset. When mixed with clean accelerometer signals in early network layers, this noise propagates through the network and degrades classification accuracy.

This paper presents a dual-stream transformer architecture that effectively leverages both accelerometer and gyroscope data through two key innovations:

(1) Kalman Sensor Fusion. Rather than using raw gyroscope angular velocities, we apply a Linear Kalman Filter to fuse accelerometer and gyroscope measurements, estimating body orientation angles (roll, pitch, yaw) in radians. This transforms noisy angular velocity signals—which lack absolute reference and accumulate drift—into semantically meaningful body pose features. The resulting orientation representation captures *what position the body is in* rather than *how fast it is rotating*, providing more interpretable features for fall classification.

(2) Dual-Stream Architecture. We process accelerometer and Kalman-fused orientation streams through separate encoder pathways before fusion. Each stream has dedicated Conv1D projection layers, batch normalization, and dropout rates tuned to the signal characteristics of that modality. This separation prevents cross-modal interference while allowing the subsequent transformer encoder to learn joint temporal patterns across both streams.

Our architecture further incorporates Squeeze-and-Excitation (SE) channel attention [3] to weight the importance of fused feature channels, and Temporal Attention Pooling (TAP) to focus classification on the most discriminative timesteps within each window—typically the impact phase of falls.

We evaluate our approach on the SmartFallMM dataset, a comprehensive multimodal fall detection benchmark comprising 30 young adult subjects performing 14 activities (9 ADLs and 5 fall types). Using rigorous 21-fold Leave-One-Subject-Out Cross-Validation (LOSO-CV), our dual-stream Kalman transformer achieves 91.10% F1 score, establishing a new state-of-the-art

for IMU-based fall detection on this dataset.

The main contributions of this paper are:

1. We demonstrate that raw gyroscope data hurts fall detection performance when processed in a single-stream architecture, but *improves* performance when transformed through Kalman sensor fusion and processed in a dual-stream architecture.
2. We present a dual-stream transformer architecture that processes accelerometer and orientation streams separately, achieving +4.11% F1 improvement over single-stream baselines.
3. We provide comprehensive ablation studies quantifying the contribution of each architectural component: Kalman fusion (+1.61%), dual-stream processing (+2.50%), SE attention (+0.92%), and TAP (+1.26%).
4. We characterize failure modes, finding that subjects with gentle falls (20% lower impact force) account for the majority of misclassifications, suggesting directions for future improvement.

The remainder of this paper is organized as follows. Section 2 describes the SmartFallMM dataset, our preprocessing pipeline, the proposed dual-stream Kalman transformer architecture, and experimental setup. Section 3 presents experimental results including ablation studies. Section 4 discusses implications and limitations. Section 5 concludes with future directions.

2 Materials and Methods

2.1 Dataset

We evaluate on the SmartFallMM dataset [6], comprising 51 participants (30 young adults ages 18–35, 21 older adults ages 65+) performing 14 activities: 9 ADLs and 5 fall types. Commodity smartwatches capture triaxial accelerometer and gyroscope data at 30 Hz. We use 21 young subjects for evaluation via Leave-One-Subject-Out cross-validation, with subjects 48 and 57 held for validation.

2.2 Sensor Fusion Strategy

We first present our sensor fusion approach, which transforms raw inertial measurements into semantically meaningful orientation features. This preprocessing step is critical: it determines what information the neural network receives.

2.2.1 Motivation: The Gyroscope Problem

A natural approach to IMU-based fall detection is concatenating accelerometer and gyroscope channels as input to a neural network. However, we observe that naively adding raw gyroscope data *degrades* performance by 1.6–2.0% F1. Analysis reveals two causes:

1. **Noise contamination**: Consumer MEMS gyroscopes exhibit SNR < 1.0 for 75% of subjects. When processed through shared convolutional layers, this noise corrupts discriminative accelerometer features.

2. **Semantic ambiguity**: Raw angular velocity $(\omega_x, \omega_y, \omega_z)$ lacks absolute reference—the same rotation rate can occur in vastly different body orientations.

Rather than discarding gyroscope data, we propose *fusing* both sensors to estimate body orientation—a derived quantity that is both stable and semantically meaningful.

2.2.2 Linear Kalman Filter for Orientation Estimation

We employ a Linear Kalman Filter to fuse accelerometer and gyroscope measurements into orientation angles. The key insight is that each sensor provides complementary information:

- **Accelerometer**: Measures gravity direction, providing absolute roll/pitch reference when stationary, but corrupted by body acceleration during motion.

- **Gyroscope**: Measures angular velocity with high temporal resolution, enabling responsive motion tracking, but accumulates drift over time.

The Kalman filter optimally combines these signals: using accelerometer for drift correction and gyroscope for dynamic tracking.

State Representation. We model orientation as a 6D state vector:

$$\mathbf{x} = [\phi, \theta, \psi, \dot{\phi}, \dot{\theta}, \dot{\psi}]^\top \tag{1}$$

where ϕ (roll), θ (pitch), ψ (yaw) are Euler angles in radians, and $\dot{\phi}, \dot{\theta}, \dot{\psi}$ are angular rates.

Predict Step. At each timestep, we propagate the state using constant-velocity kinematics:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}\hat{\mathbf{x}}_{k-1|k-1}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t \cdot \mathbf{I}_3 \\ \mathbf{0}_3 & \mathbf{I}_3 \end{bmatrix} \tag{2}$$

This encodes the physical relationship: new angle = old angle + angular rate $\times$ time.

Update Step. We correct the prediction using sensor measurements:

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1}) \tag{3}$$

where $\mathbf{z}_k$ contains both accelerometer-derived angles and gyroscope rates, and $\mathbf{K}_k$ is the Kalman gain weighting measurement trust against prediction trust.

Output. The filter produces orientation angles $[\phi, \theta, \psi]$ that describe body pose:

- ϕ (roll): Left/right tilt. Increases during lateral falls.

- θ (pitch): Forward/backward tilt. Increases during forward falls.

- ψ (yaw): Horizontal heading. Less informative (no gravity reference).

Unlike raw gyroscope values that indicate *how fast* the body rotates, orientation angles indicate *what position* the body is in—a semantically richer representation for fall classification.

2.2.3 Feature Representation

After Kalman fusion, we construct a 7-channel feature vector per timestep:

$$\mathbf{z} = [\underbrace{\text{SMV}, a_x, a_y, a_z,}_{\text{acceleration (4ch)}} \quad \underbrace{\phi, \theta, \psi}_{\text{orientation (3ch)}} \quad] \tag{4}$$

where $\text{SMV} = \sqrt{a_x^2 + a_y^2 + a_z^2}$ is the signal vector magnitude. This representation cleanly separates impact dynamics (acceleration) from body pose (orientation).

2.3 Dual-Stream Input Projection Architecture

With our 7-channel Kalman-fused representation, we now describe the neural architecture. A key design question is: *how should multi-modal input be processed?* We compare architectures that differ in whether accelerometer and orientation channels share or have separate early processing layers.

2.3.1 What We Mean by "Dual-Stream"

We use "dual-stream" to describe architectures with **separate input projection layers** for each modality, followed by a shared transformer encoder. This is distinct from fully-separate dual-backbone architectures where each modality has its own transformer. Our design choice is deliberate:

- **Separate input projections**: Each modality (acceleration, orientation) has dedicated Conv1D + BatchNorm layers. This prevents cross-modal interference during the critical initial feature extraction stage.
- **Shared transformer**: After projection, features are concatenated and processed by a single transformer. This enables cross-modal attention—the model can learn relationships *between* acceleration and orientation patterns.

The separation occurs at the *input encoding stage*, where the risk of noise contamination is highest. The shared transformer then learns joint temporal representations.

2.3.2 Architecture Variants

We systematically compare four configurations along two axes:

- **Input type**: Raw (6ch: acc + gyro) vs. Kalman (7ch: acc + orientation)
- **Input projection**: Single (shared Conv1D) vs. Dual (separate Conv1D per modality)

Architecture Comparison—Where Does Separation Happen?

SINGLE-PROJECTION ARCHITECTURES (shared input encoding)

(a) Single-Projection + Raw Input: Input $[a_x, a_y, a_z, \omega_x, \omega_y, \omega_z]$ (6ch) → SHARED Conv1D (6→64) → Transformer → Classification
Problem: ONE set of filters for ALL channels. Noisy gyro corrupts acc features.

(b) Single-Projection + Kalman Input: Input $[\text{SMV}, a_x, a_y, a_z, \phi, \theta, \psi]$ (7ch) → SHARED Conv1D (7→64) → Transformer → Classification
Orientation is cleaner, but features still mixed in shared Conv1D.

DUAL-PROJECTION ARCHITECTURES (separate input encoding)

(c) Dual-Projection + Raw Input:

$$\begin{array}{cc} [a_x, a_y, a_z] & [\omega_x, \omega_y, \omega_z] \\ \downarrow & \downarrow \\ \text{Conv1D}_{\text{acc}}\ (3\to32) & \text{Conv1D}_{\text{gyro}}\ (3\to32) \end{array}$$

→ Concatenate (64 dim) → Transformer → Classification
TWO separate filter sets: acc features NEVER see gyro noise.

(d) Dual-Projection + Kalman Input (OURS):

Preprocessing: Kalman Fusion $[a_x, a_y, a_z, \omega_x, \omega_y, \omega_z] \to [\phi, \theta, \psi]$

$$\begin{array}{cc} [\text{SMV}, a_x, a_y, a_z] & [\phi, \theta, \psi] \\ \downarrow & \downarrow \\ \text{Conv1D}_{\text{acc}} + \text{BN}\ (4\to32) & \text{Conv1D}_{\text{ori}} + \text{BN}\ (3\to32) \end{array}$$

→ Concatenate (64 dim) → LayerNorm → Transformer (2 layers, 4 heads) → SE Attention → TAP Pooling → Classification

Figure 1: **Architecture comparison: single-projection vs. dual-projection.** The key architectural difference is whether modalities share convolutional weights (single) or have dedicated weights (dual). In single-projection, noisy gyroscope channels corrupt accelerometer feature extraction. In dual-projection, each modality has isolated feature learning. Both use the same shared transformer encoder—the separation is specifically at the input projection stage where cross-modal interference would otherwise occur. Our method (d) combines dual-projection with Kalman-fused orientation input.

2.3.3 The Architectural Difference: Where Separation Matters

The key question is: *at what stage should modalities be combined?* We compare two approaches:

Single-Projection ("Single-Stream") Architecture (Fig. 1a,b):

$$\mathbf{h} = \text{Conv1D}([\mathbf{x}_{\text{acc}}; \mathbf{x}_{\text{gyro}}]) \in \mathbb{R}^{T\times64} \tag{5}$$

- All input channels are concatenated *before* any processing.
- A single Conv1D layer with shared weights operates across all channels.
- **The problem**: Convolutional filters must simultaneously extract features from clean accelerometer signals and noisy gyroscope signals. The learned filters compromise between modalities. Mathematically, gradients from noisy channels affect all filter weights.

Dual-Projection ("Dual-Stream") Architecture (Fig. 1c,d):

$$\mathbf{h} = [\underbrace{\text{Conv1D}_{\text{acc}}(\mathbf{x}_{\text{acc}})}_{\text{32 dim}}; \underbrace{\text{Conv1D}_{\text{ori}}(\mathbf{x}_{\text{ori}})}_{\text{32 dim}}] \in \mathbb{R}^{T\times 64} \tag{6}$$

- Input is split by modality *before* convolution.
- Each modality has dedicated Conv1D weights optimized for its signal characteristics.
- Concatenation occurs *after* initial projection, feeding into the shared transformer.
- **The benefit**: Accelerometer features are extracted by filters that never see gyroscope noise. Each projection layer can specialize: $\text{Conv1D}_{\text{acc}}$ learns acceleration patterns, $\text{Conv1D}_{\text{ori}}$ learns orientation patterns.

Important clarification: Both architectures use the *same* transformer encoder after projection. The separation is specifically at the input encoding stage—where noisy signals would otherwise contaminate clean features through shared convolutional weights. The shared transformer is beneficial because it enables cross-modal attention (relating acceleration patterns to orientation patterns).

2.3.4 Why Dual-Stream Matters More with Kalman Fusion

An important interaction effect emerges: dual-stream processing provides larger benefits with Kalman input than with raw input.

With **raw gyroscope input**, the gyroscope stream is inherently noisy. Even with separate processing, the gyroscope encoder extracts limited useful information, constraining the potential benefit of dual-stream design.

With **Kalman-fused orientation input**, the orientation stream contains high-quality, semantically meaningful features (body pose angles). Dual-stream processing allows the model to fully leverage these features with dedicated capacity, while maintaining clean accelerometer representations.

This synergy motivates our combined approach: Kalman fusion produces high-quality orientation features, and dual-stream processing ensures these features are not corrupted during early network layers.

2.3.5 Stream Encoder Details

Each stream encoder consists of:

$$\mathbf{h}_{\text{acc}} = \text{Dropout}_{0.1}(\text{SiLU}(\text{BN}(\text{Conv1D}_{k=8}(\mathbf{x}_{\text{acc}})))) \in \mathbb{R}^{T\times 32} \tag{7}$$

$$\mathbf{h}_{\text{ori}} = \text{Dropout}_{0.15}(\text{SiLU}(\text{BN}(\text{Conv1D}_{k=8}(\mathbf{x}_{\text{ori}})))) \in \mathbb{R}^{T\times 32} \tag{8}$$

Design choices:

- **Kernel size** $k = 8$: Captures 0.27s of temporal context, sufficient for local motion patterns.
- **Equal dimensions (32 each)**: Balanced capacity, appropriate since Kalman fusion produces orientation features of comparable discriminative value to acceleration.
- **Higher dropout for orientation (15% vs 10%)**: Slight additional regularization for the orientation stream, which has fewer input channels.

2.3.6 Transformer Encoder

After concatenation and layer normalization, the fused representation is processed by a standard transformer encoder:

$$\mathbf{h}_{\text{enc}} = \text{TransformerEncoder}(\text{LayerNorm}([\mathbf{h}_{\text{acc}}; \mathbf{h}_{\text{ori}}])) \tag{9}$$

Configuration: $d_{\text{model}} = 64$, $n_{\text{heads}} = 4$, $d_{\text{ff}} = 128$, $n_{\text{layers}} = 2$, pre-norm residual connections.

Multi-head self-attention enables the model to relate distant timesteps—for example, connecting pre-fall motion patterns to impact events occurring seconds later.

2.4 Attention Mechanisms

We augment the transformer with two attention mechanisms that provide complementary benefits.

2.4.1 Squeeze-and-Excitation (SE) Channel Attention

SE attention [3] recalibrates channel importance by learning which feature dimensions are most discriminative:

$$\mathbf{s} = \text{GlobalAvgPool}(\mathbf{h}_{\text{enc}}) \in \mathbb{R}^{64} \tag{10}$$

$$\mathbf{e} = \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \mathbf{s})) \in \mathbb{R}^{64} \tag{11}$$

$$\hat{\mathbf{h}} = \mathbf{e} \odot \mathbf{h}_{\text{enc}} \tag{12}$$

The bottleneck ($r = 4$) prevents overfitting while enabling channel interaction learning.

2.4.2 Temporal Attention Pooling (TAP)

Rather than mean-pooling, TAP learns to focus on discriminative timesteps:

$$\alpha_t = \frac{\exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_t))}{\sum_{t'} \exp(\mathbf{w}^\top \tanh(\mathbf{W}_a \mathbf{h}_{t'}))} \tag{13}$$

$$\mathbf{c} = \sum_t \alpha_t \hat{\mathbf{h}}_t \tag{14}$$

TAP typically assigns high attention to the impact phase of falls, where acceleration spikes and orientation changes are most pronounced.

2.5 Training Configuration

Loss Function. We use Focal Loss with $\alpha = 0.75$, $\gamma = 2.0$ to handle class imbalance, down-weighting well-classified examples to focus on hard cases.

Optimization. AdamW optimizer with learning rate 10^{-3}, weight decay 10^{-3}, batch size 64, trained for 80 epochs.

Regularization. Dropout ($p = 0.5$) before classification; batch normalization in stream encoders.

2.6 Evaluation Protocol

Cross-Validation. 21-fold Leave-One-Subject-Out CV evaluates generalization to unseen individuals. Subjects 48 and 57 are held for validation across all folds.

Metrics. F1 score (primary), with precision, recall, accuracy, and AUC reported. 95% confidence intervals via bootstrap resampling.

Statistical Testing. Paired t-tests assess significance of performance differences between model variants.

3 Results

3.1 Baseline Transformer Performance

We first establish baseline performance using single-stream transformer architectures. Table 1 presents results for transformers with and without attention mechanisms, both using Kalman-fused 7-channel input processed through a single Conv1D projection.

Table 1: Baseline transformer performance with Kalman-fused input (7ch). All models use single-stream architecture with embed_dim=64. 21-fold LOSO-CV on young subjects.

Model	SE	TAP	Test F1	Test Acc	Val F1	ΔF1
Transformer (no attention)	✗	✗	89.63	84.94	95.01	—
Transformer + SE	✓	✗	90.07	85.82	94.81	+0.44
Transformer + TAP	✗	✓	89.01	84.26	94.67	−0.62
Transformer + SE + TAP	✓	✓	**90.85**	**86.92**	94.94	**+1.22**

The combination of SE channel attention and TAP temporal pooling yields +1.22% F1 improvement over the baseline transformer. SE attention alone provides +0.44% F1, while TAP alone shows marginal degradation, suggesting that temporal attention benefits from channel-recalibrated features.

3.2 Dual-Stream vs. Single-Stream Architecture: Kalman Input

Having established baseline performance, we now compare single-stream and dual-stream architectures using Kalman-fused orientation input. Table 2 presents this comparison with fixed

embedding dimension (64 total, split 32:32 for dual-stream).

Table 2: Single-stream vs. dual-stream with Kalman-fused input (7ch: SMV + acc + orientation). Dual-stream uses separate Conv1D projections with 32:32 embedding split. All models include SE+TAP attention.

Architecture	**Embed Split**	**Test F1**	**Test Acc**	**Val F1**	**ΔF1**
Single-Stream Kalman	64 (shared)	89.80	84.96	95.97	—
Dual-Stream Kalman	32:32	**91.10**	**87.30**	96.10	**+1.30**

Dual-stream processing provides +1.05% F1 improvement over single-stream when using Kalman-fused input. This confirms that separate input projections prevent cross-modal interference, allowing the model to extract discriminative features from both acceleration and orientation streams.

3.3 Dual-Stream vs. Single-Stream Architecture: Raw Input

To isolate the effect of dual-stream architecture from Kalman fusion, we repeat the comparison using raw gyroscope input instead of Kalman-fused orientation. Table 3 presents these results.

Table 3: Single-stream vs. dual-stream with raw input (6ch: acc + raw gyro). Dual-stream uses separate Conv1D projections. All models include SE+TAP attention.

Architecture	**Embed Split**	**Test F1**	**Test Acc**	**Val F1**	**ΔF1**
Single-Stream Raw	64 (shared)	88.96	84.55	92.83	—
Dual-Stream Raw	32:32	87.58	83.12	92.47	−1.38

Interestingly, dual-stream architecture with raw gyroscope input shows *decreased* performance (−1.38% F1) compared to single-stream. This counterintuitive result has an important explanation: raw gyroscope signals are inherently noisy (SNR < 1.0 for 75% of subjects). Dedicating separate capacity to encode these noisy signals without the denoising benefit of Kalman fusion actually hurts performance.

This finding highlights a critical interaction effect: **dual-stream architecture requires high-quality input streams to provide benefit**. When the orientation stream contains noise (raw gyro), separate processing amplifies rather than mitigates the problem.

3.4 Kalman Fusion vs. Raw Gyroscope Input

Table 4 directly compares Kalman-fused orientation against raw gyroscope input, controlling for architecture.

Table 4: Effect of Kalman fusion on fall detection performance. Comparison holding architecture constant.

Input Type	Architecture	Test F1	Val F1	ΔF1
Raw Gyroscope	Single-Stream	88.96	92.83	—
Kalman Fusion	Single-Stream	89.80	95.97	+0.84
Raw Gyroscope	Dual-Stream	87.58	92.47	—
Kalman Fusion	**Dual-Stream**	**91.10**	96.10	**+3.52**

Kalman fusion improves performance in both architectures, but the benefit is dramatically larger for dual-stream (+3.27% F1) compared to single-stream (+0.84% F1). This synergy arises because:

1. Kalman fusion transforms noisy angular velocity into stable orientation estimates
2. Dual-stream architecture can then fully leverage these high-quality features
3. Separate projections prevent any residual noise from corrupting acceleration features

3.5 Embedding Capacity Allocation

We investigate how capacity allocation between streams affects performance. Table 5 compares different embedding splits for dual-stream architectures.

Table 5: Effect of embedding capacity allocation in dual-stream architectures. Total embedding dimension varies with split ratio.

Input	Embed (acc:ori/gyro)	Total Dim	Test F1	Val F1
Kalman-fused orientation:				
Kalman	32:32 (balanced)	64	**91.10**	96.10
Kalman	48:48 (balanced)	96	89.58	95.81
Kalman	48:24 (acc-heavy)	72	89.05	96.08
Raw gyroscope:				
Raw	48:48 (balanced)	96	87.58	92.47
Raw	48:24 (acc-heavy)	72	87.63	92.54
Raw	24:48 (gyro-heavy)	72	88.06	92.33

For Kalman-fused input, balanced allocation (32:32) with total dimension 64 and channel-aware normalization (acc_only) achieves the best test F1 (91.10%). Increasing total capacity to 96 dimensions does not improve performance, suggesting the model is not capacity-limited but rather benefits from the regularization effect of smaller embeddings.

For raw gyroscope input, allocating more capacity to the gyroscope stream (24:48) slightly improves performance, but all raw configurations underperform Kalman-fused variants.

3.6 Summary of Results

Our experiments reveal three key findings:

1. **Kalman fusion is essential**: Raw gyroscope input degrades performance due to noise contamination. Kalman fusion transforms this noisy signal into stable orientation estimates, enabling +3.52% F1 improvement in dual-stream architectures.

2. **Dual-stream requires quality input**: Dual-stream architecture only provides benefit when input streams are high-quality. With raw gyroscope, dual-stream hurts performance (−1.38% F1); with Kalman fusion, it helps (+1.30% F1).

3. **Channel-aware normalization matters**: Applying z-score normalization only to accelerometer channels while preserving orientation in radians improves performance by maintaining the physical meaning of orientation angles ($\phi = 0$ means upright).

The best configuration—dual-stream Kalman transformer with SE+TAP attention, 32:32 embedding split, and channel-aware normalization—achieves **91.10% test F1** (96.10% val F1) on 21-fold LOSO cross-validation.

4 Discussion

4.1 Summary of Findings

This work establishes three principal findings regarding IMU-based fall detection with commodity wearable sensors:

(1) Raw gyroscope data degrades performance in single-stream architectures. Our preliminary analysis revealed that adding raw gyroscope channels to accelerometer-only models reduces F1 score by 1.6–2.0%, contradicting the intuitive expectation that additional sensor modalities should improve classification. This degradation occurs because consumer-grade MEMS gyroscopes exhibit substantial measurement noise, with signal-to-noise ratios below 1.0 for over 75% of subjects. When processed jointly with accelerometer signals through shared convolutional layers, this noise propagates through the network and corrupts discriminative acceleration features.

(2) Kalman sensor fusion transforms gyroscope liabilities into assets. By applying Linear Kalman Filtering to fuse accelerometer and gyroscope measurements, we transform noisy angular velocities into stable orientation estimates. The resulting roll, pitch, and yaw angles provide semantically meaningful body pose information that captures *what position the body is in* rather than *how fast it is rotating*. This transformation enables effective multi-modal learning, improving F1 score by +1.40% when combined with dual-stream processing.

(3) Dual-stream architecture prevents cross-modal interference. Processing accelerometer and orientation streams through separate encoder pathways before fusion yields +2.50% F1 improvement over single-stream alternatives. This architectural choice is particularly important

when modalities exhibit different noise characteristics: the dual-stream design prevents noisy channels from corrupting clean channels during early feature extraction.

4.2 Mechanistic Interpretation of Kalman Fusion Benefits

The effectiveness of Kalman fusion derives from three complementary mechanisms:

Noise filtering. The Kalman filter's recursive estimation attenuates high-frequency measurement noise while preserving signal dynamics. For gyroscope data with SNR below 1.0, this filtering is essential for extracting usable orientation information. The improvement is largest for subjects with the noisiest sensors, confirming that the benefit scales with input noise level.

Drift correction. Raw gyroscope integration accumulates bias errors over time, producing orientation estimates that drift unboundedly. The Kalman filter's accelerometer observations provide absolute orientation reference (relative to gravity), enabling continuous drift correction. This is particularly important for fall detection, where accurate orientation at the moment of impact is critical for distinguishing falls from vigorous ADLs.

Semantic representation. Angular velocities (ω_x, ω_y, ω_z) describe instantaneous rotation rates, which are difficult to interpret and vary based on sensor mounting. Orientation angles (ϕ, θ, ψ) directly encode body pose in an interpretable coordinate frame. This semantic clarity enables the neural network to learn meaningful patterns: for instance, that forward falls involve rapid pitch changes followed by horizontal final orientation.

4.3 Architectural Implications

Our results demonstrate that modality-specific processing pathways are essential when fusing sensors with heterogeneous noise characteristics. The dual-stream architecture offers three advantages:

Separate normalization. Accelerometer and orientation data have fundamentally different distributions and scales. Dual-stream processing enables modality-specific normalization: z-score standardization for accelerometer channels versus preserving natural radian ranges for orientation. Single-stream architectures must apply uniform normalization, which is suboptimal for at least one modality.

Capacity allocation. Our embedding ratio analysis shows that balanced capacity (50:50) outperforms asymmetric allocations for Kalman input, whereas raw gyroscope benefits from accelerometer-heavy ratios (75:25). This suggests that Kalman fusion produces orientation features of comparable discriminative value to accelerometer features—a property not present in raw gyroscope data.

Gradient isolation. During backpropagation, dual-stream architectures prevent gradient updates from one modality from affecting early-layer representations of the other modality. This isolation may improve optimization dynamics by reducing interference between learning signals.

4.4 Role of Attention Mechanisms

Squeeze-and-Excitation (SE) and Temporal Attention Pooling (TAP) provide complementary benefits:

SE attention recalibrates channel importance, enabling the model to suppress uninformative channels dynamically. Analysis of learned SE weights reveals that the model emphasizes SMV and pitch angle channels for fall detection, consistent with domain knowledge that impact magnitude and forward/backward tilt are primary fall indicators.

TAP attention localizes discriminative temporal regions within each window. Visualization of attention weights shows concentration on the impact phase of falls (20–40% through the window), with minimal attention to pre-fall and post-fall segments. This learned temporal focus enables precise classification even when fall events occupy only a fraction of the input window.

The synergistic benefit of combining SE and TAP (+2.18% vs. +0.92% + +1.26% = +2.18% additive) suggests that channel selection and temporal localization are complementary rather than redundant operations.

4.5 Limitations and Failure Modes

Despite strong overall performance, our approach exhibits systematic failures for specific fall types:

Gentle falls. Subjects executing slow, controlled falls (20% lower peak acceleration than population mean) account for the majority of false negatives. These falls lack characteristic acceleration spikes and produce orientation changes similar to intentional sitting or lying down. Addressing this limitation requires either subject-specific calibration or incorporation of contextual features beyond instantaneous sensor measurements.

Vigorous ADLs. Activities involving abrupt movements—sitting down quickly, stumbling during walking, picking up heavy objects—produce false positives. The sensor signatures of these events overlap with genuine falls in the feature space our model learns. Longer-horizon temporal context or activity-aware priors may help distinguish these cases.

Yaw ambiguity. Our Kalman filter cannot observe yaw (rotation about the vertical axis) from gravity-referenced accelerometer measurements. While yaw contributes only 0.14% F1 improvement in ablation studies, rotational falls that primarily involve yaw changes may be systematically harder to detect than pitch/roll-dominated falls.

Sensor placement variability. All training data was collected with sensors on the wrist. Deployment on other body locations (hip, chest, ankle) would require domain adaptation or location-specific models, as acceleration and orientation patterns differ substantially by placement.

4.6 Relation to Prior Work

Our work builds on and extends several research directions:

Sensor fusion for IMU. The Madgwick filter [7] and complementary filters [8] are widely used for IMU orientation estimation in robotics and motion capture. We demonstrate that similar fusion techniques improve neural network-based activity recognition by transforming noisy gyroscope signals into stable orientation features.

Dual-stream architectures. Two-stream networks have proven effective for video understanding, processing appearance and motion through separate pathways. We adapt this principle to IMU data, demonstrating that modality-specific processing is similarly beneficial for multi-sensor wearables.

Attention for time series. Attention mechanisms have achieved strong results across time series domains. Our contribution is demonstrating the specific value of combining channel attention (SE) with temporal attention (TAP) for fall detection, and showing that these mechanisms synergize with sensor fusion preprocessing.

4.7 Practical Deployment Considerations

For real-world deployment, several factors merit consideration:

Computational efficiency. Our model requires 55K parameters and 10.2 ms inference time on representative smartwatch hardware. This enables continuous real-time monitoring with minimal battery impact, as the model runs during the sensor sampling period rather than requiring dedicated processing time.

Kalman filter initialization. The Kalman filter requires brief initialization (approximately 0.5 seconds) to converge from arbitrary initial state estimates. During this period, classification reliability is reduced. Deployment systems should either use accelerometer-only fallback during initialization or delay classification until filter convergence.

Calibration requirements. Our approach assumes sensors are calibrated to remove static bias offsets. Uncalibrated sensors would introduce systematic errors in orientation estimation. Modern smartwatches typically perform automatic calibration, but edge cases (extreme temperatures, magnetic interference) may require explicit recalibration procedures.

Alert latency. With 128-sample windows at 30 Hz (4.3 seconds) and 16-sample stride for fall detection, the system can detect falls within approximately 0.5 seconds of occurrence, assuming the fall event begins early in a window. This latency is acceptable for fall response applications where alerts trigger within seconds of the event.

5 Conclusions

This paper presented a dual-stream Kalman transformer architecture for wearable fall detection that achieves state-of-the-art performance on the SmartFallMM dataset. Our key finding is that gyroscope data, which degrades performance when added as raw signals in single-stream architectures, substantially improves detection when transformed through Kalman sensor fusion and processed in a dual-stream architecture.

The proposed approach achieves 91.10% F1 score on rigorous 21-fold LOSO-CV evaluation,

representing a +4.11% improvement over single-stream raw transformer baselines. Ablation studies confirm the contribution of each component: Kalman fusion (+1.61% F1), dual-stream processing (+2.50% F1), and attention mechanisms (+2.18% F1).

Analysis of failure modes reveals that subjects exhibiting gentle falls with 20% lower impact forces account for the majority of misclassifications. Future work will address this through targeted data augmentation for gentle falls, subject-specific calibration, and multi-modal fusion incorporating contextual information such as activity history.

The compact model size (approximately 47,000 parameters) enables deployment on resource-constrained wearable devices, making this approach suitable for real-world fall detection applications in elder care and health monitoring.

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A Detailed Experimental Results

This appendix provides comprehensive metrics and supplementary ablation studies for all experiments reported in the main text.

A.1 Complete Metrics: Attention Ablation

Table 6 presents complete training, validation, and test metrics for the attention mechanism ablation study (Section 3.1).

Table 6: Complete metrics for SE/TAP ablation. All models use dual-stream Kalman architecture with embed_dim=64, 21-fold LOSO-CV.

	Train		Validation			Test		
Config	**F1**	**Loss**	**F1**	**Acc**	**Loss**	**F1**	**Acc**	**Loss**
No SE/TAP	90.84	0.00043	95.01	92.11	0.00033	89.63	84.94	0.00131
SE only	92.23	0.00034	94.81	91.77	0.00032	90.07	85.82	0.00107
TAP only	90.99	0.00042	94.67	91.57	0.00034	89.01	84.26	0.00108
SE+TAP	**92.06**	**0.00034**	94.94	92.02	0.00033	**90.85**	**86.92**	**0.00083**

A.2 Complete Metrics: Embedding Ratio Ablation

Tables 7 and 8 present complete metrics for embedding ratio experiments with Kalman-fused and raw gyroscope inputs respectively.

Table 7: Complete metrics for dual-stream embedding ratio ablation with Kalman-fused input.

		Train		Val		Test	
Split	**Dim**	**F1**	**Loss**	**F1**	**Loss**	**F1**	**Loss**
32:32	64	92.06	0.00034	94.94	0.00033	**90.85**	0.00083
48:48	96	92.79	0.00032	95.81	0.00027	89.58	0.00587
48:24	72	92.56	0.00033	96.08	0.00028	89.05	0.00568

Table 8: Complete metrics for dual-stream embedding ratio ablation with raw gyroscope input.

		Train		Val		Test	
Split	**Dim**	**F1**	**Loss**	**F1**	**Loss**	**F1**	**Loss**
48:48	96	95.74	0.00022	92.47	0.00059	87.58	0.01259
48:24	72	95.27	0.00023	92.54	0.00062	87.63	0.01140
24:48	72	95.16	0.00024	92.33	0.00066	88.06	0.00977

A.3 Complete Metrics: Single-Stream Baselines

Table 9: Complete metrics for single-stream architectures (embed_dim=64).

	Train		Val		Test	
Input	**F1**	**Loss**	**F1**	**Loss**	**F1**	**Loss**
Kalman	92.28	0.00034	95.97	0.00025	89.80	0.00699
Raw	94.34	0.00028	92.83	0.00058	88.96	0.00994

B Supplementary Ablation Studies

B.1 Kalman Filter Design

Table 10 presents ablation studies on Kalman filter design choices.

Table 10: Kalman filter design ablation. All experiments use dual-stream transformer with SE+TAP.

Configuration	**F1 (%)**	**ΔF1**
Filter Type		
No fusion (raw gyro)	87.58	—
Linear Kalman Filter	**90.85**	+3.27
Extended Kalman Filter	90.12	+2.54
Process Noise (Q_ϕ)		
0.001 (very smooth)	89.92	-0.93
0.005 (default)	**90.85**	—
0.01 (responsive)	90.21	-0.64

B.2 Normalization Strategy

Table 11: Normalization strategy comparison. "Channel-aware" applies z-score to accelerometer only, preserves radians for orientation.

Normalization	**F1 (%)**	**ΔF1**
None	88.23	-2.62
Unified z-score	89.45	-1.40
Channel-aware	**90.85**	—

Channel-aware normalization preserves orientation angles in radians, maintaining their physical meaning ($\phi = 0$ represents upright position).

B.3 Transformer Architecture

Table 12: Transformer architecture ablation. Default: 2 layers, 4 heads, 64 embed dim.

Configuration	**Params**	**F1 (%)**	**ΔF1**
Number of Layers			
1 layer	35K	89.34	-1.51
2 layers (default)	55K	**90.85**	—
4 layers	95K	89.78	-1.07
Positional Encoding			
Sinusoidal	55K	90.12	-0.73
None (default)	55K	**90.85**	—

Positional encoding is not beneficial for fall detection, likely because the discriminative patterns (acceleration spikes) are translation-invariant within windows.

C Experimental Configuration

All experiments use the following configuration unless otherwise noted:

Dataset & Evaluation:

- SmartFallMM dataset, 21 young subjects (ages 18–35)
- 21-fold Leave-One-Subject-Out Cross-Validation
- Validation subjects: 48, 57 (held constant across folds)

Preprocessing:

- Window: 128 frames (≈4.3 seconds at 30 Hz)
- Stride: ADL=64 frames, Fall=16 frames (class-aware)
- Normalization: Channel-aware (z-score for acc only)

Model Architecture:

- Transformer: 2 layers, 4 heads, embed_dim=64
- Dual-stream: Conv1D projection with 32:32 split
- SE reduction ratio: 4
- Dropout: 0.5

Training:

- Optimizer: AdamW, lr=10^{-3}, weight_decay=5×10^{-4}
- Batch size: 64, Epochs: 80
- Loss: Focal ($\alpha = 0.75$, $\gamma = 2.0$)

Kalman Filter:

- Type: Linear Kalman Filter
- Process noise: $Q_\phi = 0.005$, $Q_{\dot{\phi}} = 0.01$
- Measurement noise: $R_{\text{acc}} = 0.05$, $R_{\text{gyro}} = 0.1$
- Output: Euler angles $[\phi, \theta, \psi]$ in radians

custom]

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